Unit 2 Quiz: Analyzing One Variable KEY

Name _____

Show your work.

2.1 - Standard Deviation

1. **I find the standard deviation of a small data set.**

   Kim counted how many text messages she got each hour for 5 hours.

   2, 6, 0, 4, 3

   (a) Find the mean, $\bar{x}$. Then complete the table.

   $\bar{x}$ = 15 ÷ 5 = 3

x	$x - \bar{x}$	$(x - \bar{x})^2$
2	−1	1
6	3	9
0	−3	9
4	1	1
3	0	0
	$\Sigma(x - \bar{x})^2$	20

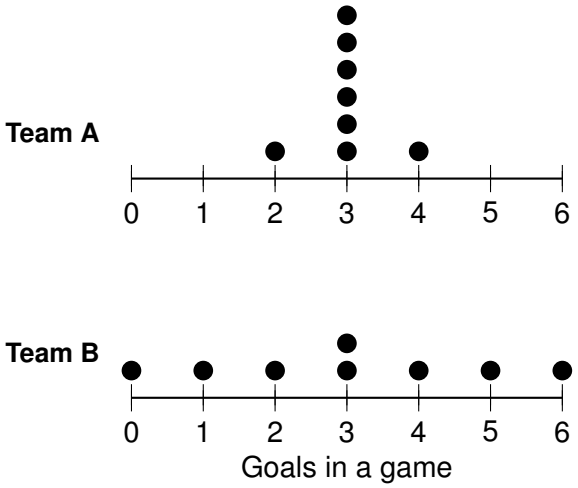
   (b) Find the standard deviation, s. Round to 2 decimal places.

   $$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}} = \sqrt{\frac{20}{4}}$$

   $= \sqrt{5} \approx 2.24$ messages

2. **I compare the spread of data sets by looking at graphs, and I explain what the standard deviation tells about data.**

   These dot plots show the goals two soccer teams scored in 8 games. Both teams have a mean of 3 goals.



   (a) Which team has the larger standard deviation? Explain.

   Team B. Its goals are spread far from 3. Team A's goals are packed close to 3.

   (b) Team B's SD is 2 goals. What does this number tell you about Team B?

   Team B's goals in a game are usually about 2 goals away from its mean of 3.

   (c) Team C scored exactly 3 goals in every game. What is Team C's SD? Explain.

   0. Every value is the same as the mean, so there is no spread.

2.2 - Choosing Measures and Comparing Distributions

3. **I choose mean and SD or median and IQR based on the shape of the data.**

   For each data set, circle the measures you should use. Then explain.

   (a) The minutes 30 students spent on the bus today. The histogram is skewed right.

   mean and SD (median and IQR) Explain: The data is skewed.

   (b) The lengths of 100 leaves from one tree. The histogram is symmetric, with no outliers.

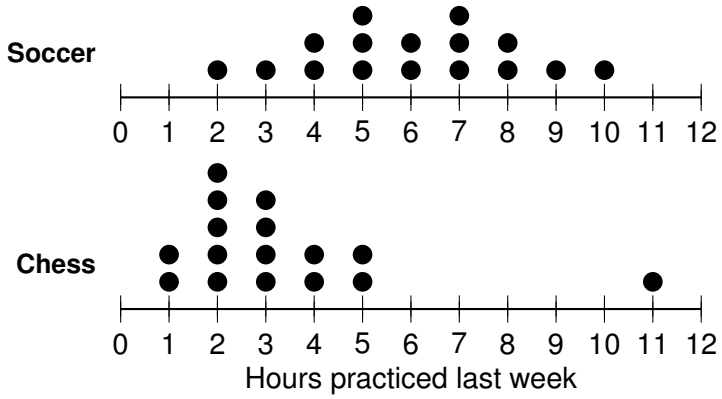
   (mean and SD) median and IQR Explain: The data is symmetric with no outliers.

   (c) The prices of 7 used phones, in dollars: 120, 140, 150, 150, 160, 180, 450

   mean and SD (median and IQR) Explain: 450 is an outlier. (Q1 = 140, Q3 = 180, IQR = 40, upper fence = 180 + 60 = 240.) It pulls the mean and the SD.

4. **I compare two distributions using shape, center, spread, and outliers.**

   These dot plots show how many hours students in two clubs practiced last week.



	Q1	Median	Q3	IQR
Soccer	4.5	6	7.5	3
Chess	2	3	4	2

   (a) Describe the shape of each club's data.

   Soccer is symmetric.
   Chess is skewed right.

   (b) Compare the centers. Use numbers and the context.

   The median for soccer (6 hours) is greater than for chess (3 hours). Soccer players usually practiced 3 hours more.

   (c) Compare the spreads. Use numbers.

   The IQR for soccer (3 hours) is greater than for chess (2 hours). The middle half of the soccer players is more spread out.

   (d) Does either club have an outlier? Explain.

   Yes. One chess player practiced 11 hours. The upper fence for chess is $4 + 1.5 \times 2 = 7$, and 11 is above 7.

2.3 - Percentiles and Z-Scores

5. **I find the percentile of a value.**

   20 students said how many minutes they played a phone game last night. Here are the times, in order.

   5, 8, 12, 15, 18, 20, 22, 25, 27, 30, 32, 35, 38, 40, 42, 45, 50, 55, 60, 75

   (a) What percentile is 45 minutes?

   15 times are below 45. $15 \div 20 = 0.75$, the 75th percentile

   (b) What percentile is 15 minutes?

   3 times are below 15. $3 \div 20 = 0.15$, the 15th percentile

   (c) Which time is at the 40th percentile?

   40% of 20 is 8 times below it. The 9th time is **27 minutes**.

6. **I find and explain a z-score, and I compare values from different data sets with z-scores.**

   Amina took two tests.

Test	Amina's score	Class mean μ	Class SD σ
Math	79	73	4
Reading	91	83	8

   (a) Find Amina's z-score on each test.

   Math: $z = \dfrac{x - \mu}{\sigma} = \dfrac{79 - 73}{4} = \dfrac{6}{4} = 1.5$ Reading: $z = \dfrac{91 - 83}{8} = \dfrac{8}{8} = 1$

   (b) What does Amina's z-score on the math test tell you?

   Her math score is 1.5 standard deviations above the class mean.

   (c) Amina's reading score is higher. On which test did she do better compared to her class? Explain.

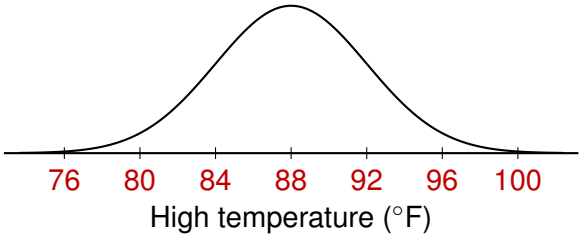
   Math. Her z-score on math (1.5) is larger than her z-score on reading (1).

2.4 - Normal Distributions and the Empirical Rule

7. **I use the empirical rule (68-95-99.7) to find percents, and I find the percentile of a value 1, 2, or 3 SDs from the mean.**

The high temperature each day in July in one city has a normal distribution. The mean is 88°F and the SD is 4°F.

(a) Label the number line with the temperatures.



(b) About what percent of days have a high temperature from 84°F to 92°F?

68% (within 1 SD)

(c) About what percent of days have a high temperature above 96°F?

2.35 + 0.15 = 2.5%

(d) About what percent of days have a high temperature from 80°F to 92°F?

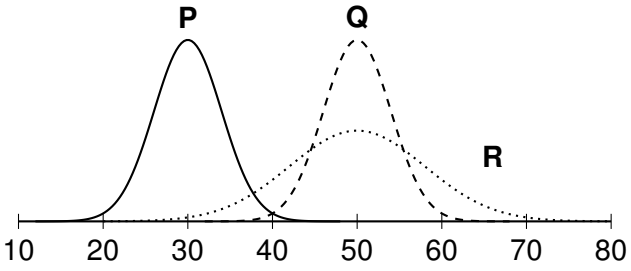
13.5 + 34 + 34 = 81.5%

(e) What percentile is a high temperature of 92°F?

92 is 1 SD above the mean.
50 + 34 = 84, the 84th percentile

8. **I describe a normal distribution, and I compare normal curves with different means and SDs.**

The three curves are normal distributions.



(a) Which two curves have the same mean?

Q and R (both 50)

(b) Which two curves have the same SD?

P and Q (same width and shape)

(c) Which curve has the largest SD? Explain.

R. It is the widest and most spread out.

(d) Which statement is true for every normal distribution? Circle the letter.

A The mean is greater than the median.
(B) The mean and the median are equal.
C The data is skewed right.
D The mean is always 0.

2.5 - Normal Calculations with a Z-Table

Use the z-table.

9. **I read a z-table.**

Find the area **below** each z-score.

(a) $z = 0.73$

.7673

(b) $z = -1.45$

.0735

Find the area **above** or **between**.

(c) above $z = 1.10$

$1 - .8643 = .1357$

(d) between $z = -1.30$ and $z = 0.40$

$.6554 - .0968 = .5586$

10. **I find the percent of data below, above, or between values in a normal distribution.**

A machine fills bottles with water. The amounts have a normal distribution with a mean of 500 mL and an SD of 6 mL. What percent of the bottles have:

(a) less than 510 mL?

$z = \frac{510 - 500}{6} = 1.67$. Table: .9525. **95.25%**

(b) more than 491 mL?

$z = \frac{491 - 500}{6} = -1.5$. Table: .0668. $1 - .0668 = .9332$. **93.32%**

(c) from 497 mL to 509 mL?

$z = \frac{497 - 500}{6} = -0.5$, table .3085. $z = \frac{509 - 500}{6} = 1.5$, table .9332.
$.9332 - .3085 = .6247$. **62.47%**

2.6 - Normal Calculations in Reverse

Use the z-table.

11. **I find the z-score for a percentile with the z-table, and I find the value at a percentile in a normal distribution.**

    (a) Find the z-score at the 60th percentile.

    Closest area .5987: $z = 0.25$

    (b) Find the z-score at the 45th percentile.

    Closest area .4483: $z = -0.13$

    Scores in a video game have a normal distribution. The mean is 1,200 points and the SD is 200 points.

    (c) What score is at the 35th percentile?

    Area .3500. Closest area .3483: $z = -0.39$

    $x = \mu + z\sigma = 1200 + (-0.39)(200) = 1200 - 78 = 1122$ points

12. **I find cutoffs for "top" and "bottom" percents.**

    The weights of a bakery's loaves of bread have a normal distribution. The mean is 680 grams and the SD is 12 grams.

    (a) The lightest 4% of the loaves are sold at a lower price. Below what weight is a loaf sold at a lower price?

    Bottom 4%: the area to the left is .0400. Closest area .0401: $z = -1.75$

    $680 + (-1.75)(12) = 680 - 21 = 659$ grams

    (b) The heaviest 30% of the loaves get a "large" sticker. How heavy must a loaf be to get the sticker?

    Top 30%: the area to the left is $1 - .30 = .70$. Closest area .6985: $z = 0.52$

    $680 + 0.52 \times 12 = 680 + 6.24 = 686.24$. About **686.2 grams**